All logos from my certs or company web sites.
Copyrights owned by company issuing the cert.

Why do employers want you certified?

Required for job role

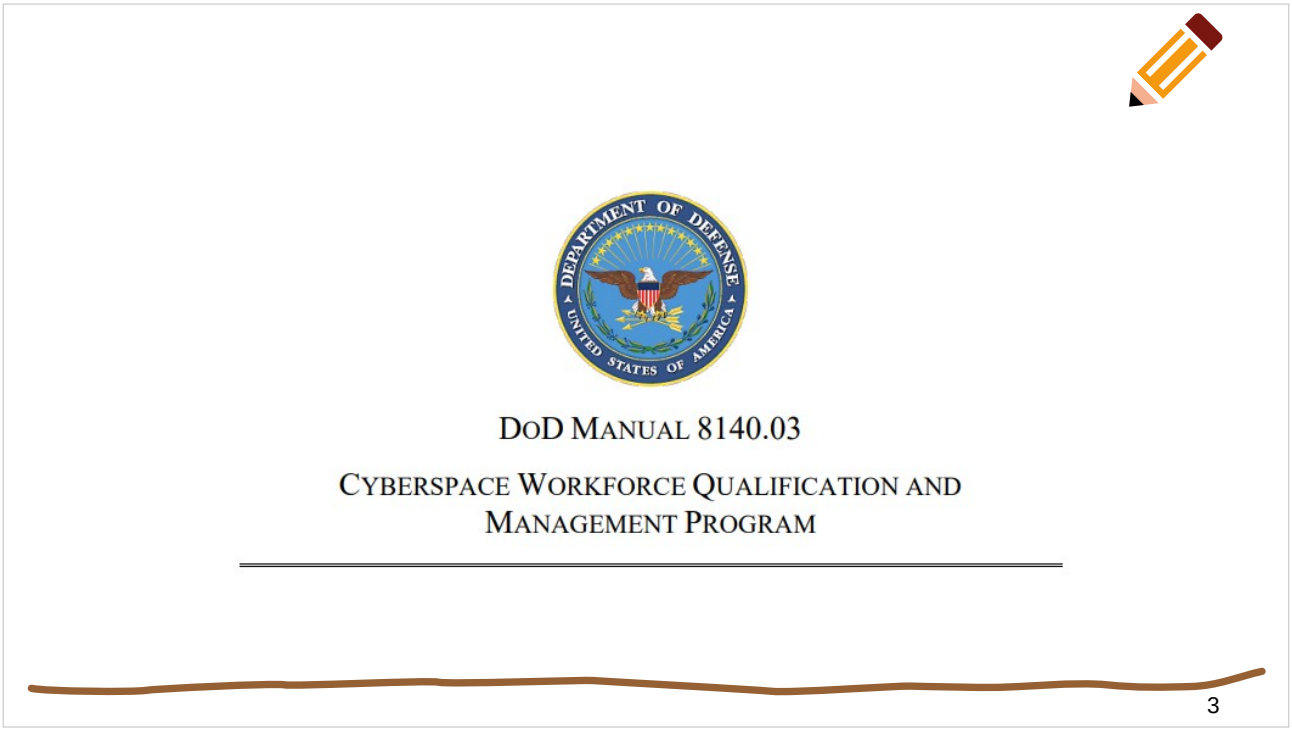
Marketing company services

Reduced outsource costs

Faster hiring and onboarding and lower turnover

Employers request certifications for a variety of reasons.

Required for job role - such as DoD Directive 8140
Marketing company services – consultants and support reps are certified in the product.
Reduced outsource costs – I have a story
Faster hiring and onboarding and lower turnover – certified individuals have a 15% longer tenure.



https://dodcio.defense.gov/Portals/0/Documents/Library/DoDM-8140-03.pdf

Cert to roles mappings:

Cisco: https://www.cisco.com/site/us/en/learn/training-certifications/training/dod-8140.html

CompTIA: https://www.comptia.org/en-us/blog/what-8140-means-for-companies-working-with-dod/

All ISC2 certs (not mapped but has nice list of roles): https://www.isc2.org/training/us-government#Cybersecurity%20Readiness%20Training%20and%20Certification%20for%20You

GIAC certs with SANS training: (more advanced roles, lists under role categories at bottom, not direct map) https://www.giac.org/workforce-development/dodd-8140/

DoD Directive 8140

- Replaced 8570 in 2023
- Mandates certification for personnel with privileged access to DoD information systems
- Applies to military personnel, government civilians, and contractors.
- Roles are categorized by 3 technical levels
- Entry level includes CompTIA A+, CCNA
- Intermediate Level includes CompTIA Security+

https://dodcio.defense.gov/Portals/0/Documents/Library/DoDM-8140-03.pdf

https://dl.dod.cyber.mil/wp-content/uploads/8140/pdf/unclass-dod8570_ia_program_transition_dod8140_cwp.pdf

2025 Value of IT Certification Candidate Report https://go.pearsonvue.com/voc

Pearson vue 2025 Value of IT Certification Candidate Report

Certifications On the Rise

93% of respondents reported having at least one certification, nearly a 6% increase from last year, and reported benefits like improved work quality (60%), engagement (48%), and productivity (43%).

Economic Impact of Training

97% of IT decision-makers say certified staff add value to their organizations, with 22% estimating that value at $30,000 or more. These professionals close skill gaps, solve problems faster, and give their employers a competitive edge.

Skill Gaps Hurt Business Outcomes

Two-thirds of decision-makers report skill gaps on their teams, with nearly as many forecasting them in the next few years. The cost of skill gaps burden teams, increasing their stress and hampering their ability to meet business objectives.

2024 report: https://www.globalknowledge.com/us-en/content/salary-report/it-skills-and-salary-report/

https://www.globalknowledge.com/us-en/content/salary-report/it-skills-and-salary-report/

SkillSoft Skillsoft’s It Skills and Salary

Taking a Certification Exam

- Preparation
  - Choose the right cert for your needs and training experience
  - There are different styles of testing
  - Study, practice, apply
- Purchasing and Scheduling
- The Day of…
- Getting your results
- Retake options

Mention confidentiality clauses

Preparation is about more than just technical knowledge
Test taking skills – multi choice is different than 100% hands-on
Mental preparation – sleep, relaxation

Choosing the right certification

- What are your goals?
- What are your Interests?
- Research your options
  - Availability, reputation, recognition
  - General or product specific
  - Test styles, costs, etc
- Think long term
  - What is the re certification process?

https://www.globalknowledge.com/us-en/resources/resource-library/articles/how-to-select-the-right-certification-for-you/

https://www.globalknowledge.com/us-en/resources/resource-library/articles/how-to-select-the-right-certification-for-you/

Styles

- Multi-guess
- Simulations
- Live Labs (Virtual Machines)
- Mixed
- 100% Hands-on

Types explained next slides

Multi-guess

- Select best answer, select all that apply, matching
- Scenarios with multiple questions
  - Performance based questions
- Watch for negatives
- Some are adaptive

- Prepare with flashcards (terminology) and practice tests
- Value can be limited by “exam dumping”

Adaptive:
Cannot go back and review previous questions/answers
A wrong answer can trigger more questions on the same objective – do you really know it or not
A correct answer can trigger harder questions on the same objective – what else do you know

Scenario (case study) questions may be performance based by asking questions about command or log output.

Value limited by “exam dumping”
AWS uses simulations “to prevent "exam dumping" (memorization) and ensure candidates can apply knowledge in real-world scenarios.”

CompTIA exams advertise as performance based exams but it is through Performance-Based Questions (PBQs) to simulate real-world tasks, such as configuring a firewall or troubleshooting a

Simulations

- Often included with multi-guess computer based exams
  - Performance based questions
- Enter full commands
- May not have all the shortcut keys of live environment
  - Tab complete
  - In app help
- Often no access to the documentation
- Prepare with practice and repetition. Some simulators are available for learning and practice.

Prepare with lots of practice. Repetition, muscle memory for common commands.
Think through what commands you may be asked to do…. Save config for persistence, enter priv mode, etc.

Cisco has used a lot of simulations for years.
CompTia, GIAC, AWS, Azure all now do as well.

Hands-on Live Labs

- May be performance based questions mixed with simulations and multi-guess (CompTia, GIAC)
- Some exams are 100% hands-on
- Example: Red Hat exams
  - Test end result, not method
  - Can use graphical or CLI tools as available
  - Red Hat even includes standard documentation and some ability to install software that is included with the product.
- If documentation is provided – know the docs!
  - What to memorize vs what to know where to find examples

Stories of Cisco Certified Internetwork Expert (CCIE) in person, multi day, break things while you are out. Late 90's

One of the designers of that went to RH to help with the RHCE

Stories of my first RHCE – on zip drives.

Now who has them?

All Red Hat exams – add link to youtube videos that show how to take

https://access.redhat.com/community/learn has the intro video….

https://www.youtube.com/watch?v=Me6Y12-sux8&t=133s

OffSec Certified Professional (OSCP) - for pen testers and ethical hackers. 24 hours live labs

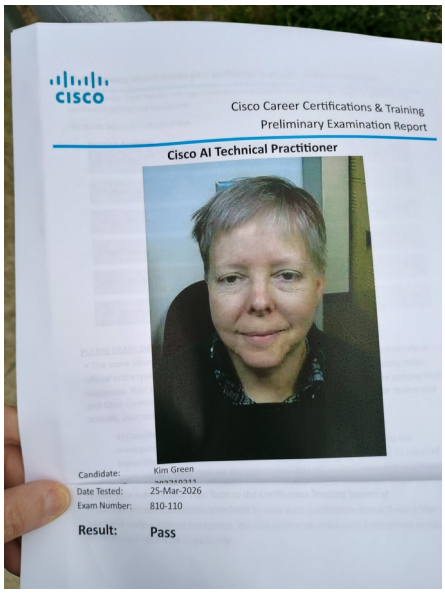
GIAC Applied Knowledge Exams – Designed for top-tier cybersecurity specialists, these 100% hands-on, CyberLive-based exams set a new standard in

The Night before to the Day of...

- Read instructions on what to bring and what not to bring.
- Be on time (early even) and have the required ID.
- Sleep can help more than last minute cramming.
- Same with relaxation exercises. Breath.
- Can be at a site or from home depending on the exam
  - Home based exams often require a camera for remote monitoring and a clean, quiet space with no other people present.
  - Test centers can be cold. Dress comfortably and in layers.

Results

- Computer based at testing site usually provide a score immediately. These are usually multi-guess and mixed styles.
- Performance based exams are often graded after and scores reported by email withing a few days.
- Pass/Fail and sometimes a number score – points or percent
- Some break down by list of topics or objectives



The results of your recent EX432 Red Hat Certified Specialist in OpenShift Advanced Cluster Management Exam are reported below.

Exam domain number: 7
Passing score: 210
Your score: 257

Result: PASS

Congratulations -- you have earned the Red Hat Certified Specialist in OpenShift Advanced Cluster Management certification.

Performance on exam objectives:

OBJECTIVE: SCORE
Manage and deploy Red Hat Advanced Cluster Management for Kubernetes (RHACM): 100%
Manage clusters by using Red Hat Advanced Cluster Management for Kubernetes (RHACM): 100%
Configure access control for multi-cluster management: 100%
Deploy and manage policies for multiple clusters: 100%
Manage the RHACM observability service: 100%
Use Kustomize to manage resources: 0%
Manage multi-cluster application lifecycle using GitOps practices: 50%
Manage Virtualization Operator: 100%

Pics of pass scores

The results of your recent EX288 Red Hat Certified OpenShift Application Developer Exam are reported below.

Exam domain number: 19
Passing score: 210
Your score: 162

Result: NO PASS

Performance on exam objectives:

OBJECTIVE: SCORE
Work with Red Hat OpenShift Container Platform: 78%
Deploy multi-container applications: 100%
Work with container images in OpenShift Container Platform: 0%
Troubleshoot application builds and deployment issues: 60%
Work with image streams: 100%
Work with configuration maps and secrets: 75%
Work with the source-to-image (S2I) framework: 17%
Work with templates: 50%
Work with OpenShift Pipelines: 20%

Please contact us within 30 days if you have any questions, concerns or suggestions about your exam or certification at

Pic of fail score

Learn from the experience

- Pass or fail, you can learn from the experience and results
- Some exam purchase options include a retake in the cost if you fail the first time

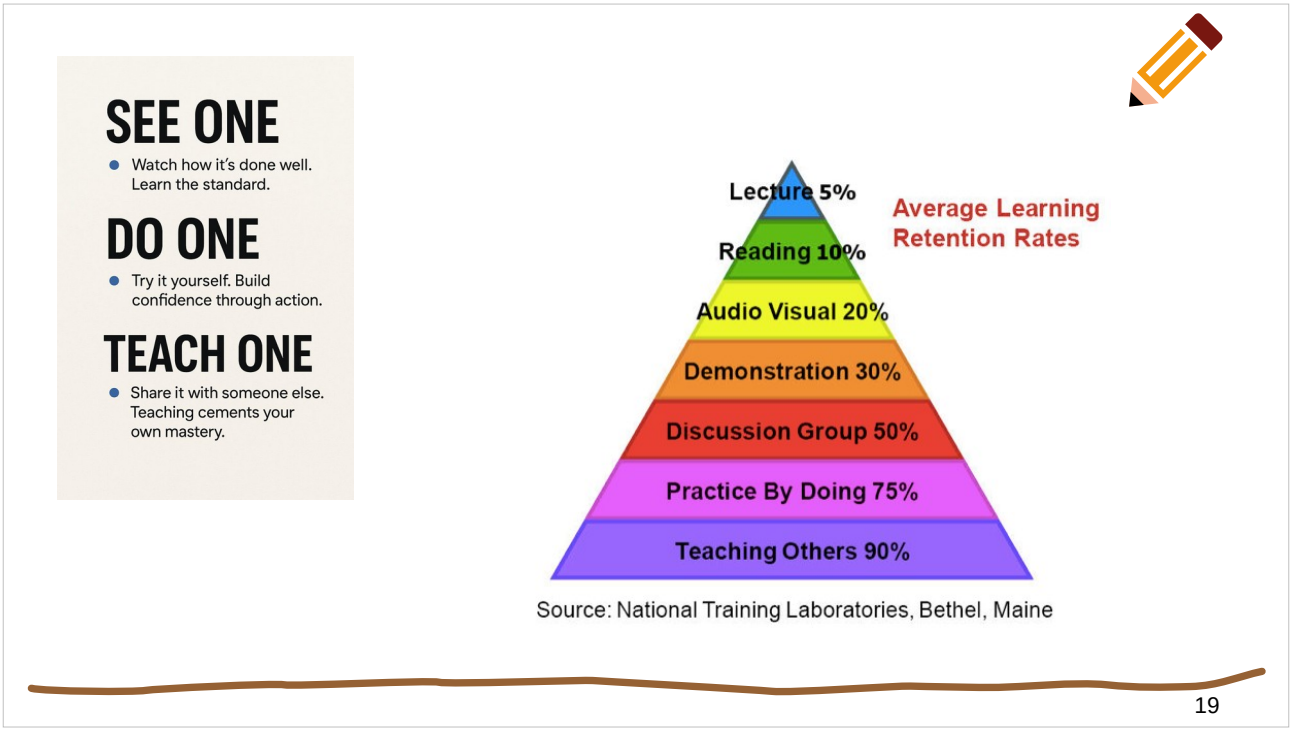
Nike logo from https://en.wikipedia.org/wiki/Just_Do_It
Yoda image from https://tenor.com/search/do-or-do-not-there-is-no-try-gifs

Study tips

- Terminology
  - Understand the question, especially for performance objectives
  - Flashcards
- Practice
  - Repeat training materials labs
  - Repetition of common commands and options
  - Looking up specifics in the documentation
- Apply
  - Work with the product outside of the training materials

Apply – how would the lab scenario be different for a different client?
What changes – name of device, IP address, etc
What stays the same – command to save config from RAM to disk for persistance

See one, do one, teach one.



How to learn – SODOTO – started in medical/surgical training in 1800's – can be see or watch

How to retain information – National Training Laboratories (NTL) – study not validated, methodology unknown, oversimplified, passive learning is still important.

Thank You!

- Susan Lauber
- sml@laubersolutions.com
- https://www.laubersolutions.com
- https://github.com/laubersm
-
- github.com/laubersm/LauberSolutions/tree/main/finalPDFs
  - PDF of slides
  - Get Certified Links text file